



Compensation relationship in thermodynamics of vaporization of aromatic and aliphatic compounds and their heat capacities

Mikhail I. Yagofarov^{*}, Boris N. Solomonov

Department of Physical Chemistry, Kazan Federal University, Kremlevskaya str. 18, 420008 Kazan, Russia

ARTICLE INFO

Keywords:

Vaporization enthalpy
vaporization Gibbs energy
Compensation relationship
Vapor pressure
Trouton's rule
Vaporization entropy
Heat capacity

ABSTRACT

It is long known that there is a correlation between the Gibbs energies and enthalpies of many processes, particularly solvation and vaporization, which is called the compensation relationship. Recently, we established that the correlation between the Gibbs energies and enthalpies of vaporization of non-hydrogen-bonded organic non-electrolytes at 298.15 K is divided into the two linear relationships with different slopes in the cases of aromatic and long-chain aliphatic compounds. We also recently found that there is a correlation between the difference between the heat capacities of the liquid and ideal gas, on the one hand, and the vaporization enthalpy at 298.15 K. Similarly with the compensation relationship, it is different in the cases of aromatic and aliphatic compounds.

In this work, our goal was to establish the factors defining the values of the coefficients of the Gibbs energy-enthalpy correlation and understand its features in aromatic and aliphatic compounds. The thermodynamic analysis enabled to independently determine the coefficients of the compensation relationship, investigate their possible boundaries, and explicitly show how they depend on the difference between the heat capacities of the liquid and ideal gas and boiling entropy. Slight variations in the boiling entropy of non-hydrogen-bonded compounds appeared to have almost no effect on the compensation relationship, while the difference in the slopes exhibited by aromatic and aliphatic molecules was due to the features of the heat capacity-enthalpy dependence.

1. Introduction

In many processes, particularly solvation and vaporization, there is a correlation between the entropy (or Gibbs energy) and enthalpy changes in a series of systems, called the compensation relationship. Features of the compensation relationship arising in various phenomena, from binding of macromolecules to micellization and sorption, have been comprehensively reviewed by Liu and Guo [1]. There is no formal thermodynamic proof, which would give a general explanation of the compensation relationship. Since the discovery [2], attempts have been made to establish its origins [1–10].

This work is a continuation of our recent studies of the correlations between the Gibbs energies and enthalpies of solvation ($\Delta_{\text{solv}}G^\circ$ vs. $\Delta_{\text{solv}}H^\circ$) and vaporization ($\Delta_1^sG^\circ$ vs. $\Delta_1^sH^\circ$) at $T_0 = 298.15$ K manifested in different classes of organic compounds [11,12] (Eq. (1):

$$\Delta_{\text{solv}}G^\circ = a\Delta_{\text{solv}}H^\circ + b \quad (1)$$

In Eq. (1) a and b are empiric coefficients. The vaporization enthalpy/Gibbs energy equals the solvation enthalpy/Gibbs energy of a compound in its own liquid phase taken with the opposite sign, so Eq. (1) holds for $-\Delta_1^sG^\circ$ and $-\Delta_1^sH^\circ$ as well. Since 1930s [2], the values of 0.6 ± 0.1 have been often reported for a and 15 ± 5 kJ·mol^{−1} for b .

Analyzing the relationship between $\Delta_{\text{solv}}G^\circ$ ($\Delta_1^sG^\circ$) and $\Delta_{\text{solv}}H^\circ$ ($\Delta_1^sH^\circ$) at $T_0 = 298.15$ K, we showed that there is a systematic difference in the slopes of Eq. (1) between non-hydrogen-bonded aromatic and long-chain aliphatic compounds [11]. It is also long known that there are systematic deviations from the common $\Delta_{\text{solv}}G^\circ$ ($\Delta_1^sG^\circ$) vs. $\Delta_{\text{solv}}H^\circ$ ($\Delta_1^sH^\circ$) correlation in hydrogen-bonded media [2,13–16]. In Ref. [12] these deviations were classified, depending on the hydrogen bonding ability of solvent and solute, and approaches for their prediction were proposed and validated.

The observed difference between aromatic and long-chain aliphatic compounds reminds of the correlations between the heat capacity ($\Delta_1^sC_{p,m}$) and enthalpy changes on vaporization at $T_0 = 298.15$ K (Eq. (2)

^{*} Corresponding author.

E-mail address: MiYagofarov@kpfu.ru (M.I. Yagofarov).

observed in our recent works [17–20]:

$$-\Delta_1^{\text{g}} C_{p,m} = c + d \cdot \Delta_1^{\text{g}} H^{\circ} \quad (2)$$

The coefficients c and d in Eq. (2) differ for aromatics and long-chain aliphatics, while alkylaromatics are described by a weighted sum. The same applies to the coefficients a and b in Eq. (1). The difference between the liquid and ideal gas phase heat capacities is defined by the contribution of the intermolecular interactions and dynamics in the liquid phase, while the vaporization enthalpy is a measure of the intermolecular interaction energy in the liquid. Therefore, Eq. (2) can be interpreted as a relationship between the intermolecular interaction contributions to the heat capacity and internal energy of the liquid. It is reasonable that the coefficients of Eq. (2) are different in aliphatics and aromatics, given that there are definite differences in the intermolecular interaction energetics between alkanes and arenes [21], and the hindered intramolecular rotations in flexible molecules are expected to additionally contribute to the liquid heat capacity, compared with an ideal gas.

The features of the $\Delta_{\text{solv}} G^{\circ}$ vs. $\Delta_{\text{solv}} H^{\circ}$ and $\Delta_1^{\text{g}} G^{\circ}$ vs. $\Delta_1^{\text{g}} H^{\circ}$ correlations are harder to explain. The known interpretations of the compensation relationship [1,9,10] discuss the origins of the linear dependence between $\Delta_{\text{solv}} G^{\circ}$ and $\Delta_{\text{solv}} H^{\circ}$ but do not predict its coefficients or bound their values. In this work, our goal was to understand the reasons for the difference between the $\Delta_1^{\text{g}} G^{\circ}$ vs. $\Delta_1^{\text{g}} H^{\circ}$ correlations for aromatic and long-chain aliphatic compounds, using the revealed $\Delta_1^{\text{g}} C_{p,m}$ vs. $\Delta_1^{\text{g}} H^{\circ}$ correlation traits. Second, based on the thermodynamic analysis, we aimed to find the limits of the coefficients of the $\Delta_1^{\text{g}} G^{\circ}$ vs. $\Delta_1^{\text{g}} H^{\circ}$ correlation and the factors that define them.

2. Methodology

2.1. Basic principles and definitions

The same definitions are held as in Ref. [11]. The standard molar enthalpy of vaporization ($\Delta_1^{\text{g}} H_A^{\circ}$) is the enthalpy change when 1 mol of liquid A is transferred to the ideal gas phase at $T_0 = 298.15$ K and $p^0 = 1$ bar. The standard molar enthalpy of solvation ($\Delta_{\text{solv}} H_{A/S}^{\circ}$) is the enthalpy change when 1 mol of an ideal gas phase solute A at $T = 298.15$ K and $p^0 = 1$ bar dissolves in a solvent S to give a solution of infinite dilution. Analogous definitions hold for the Gibbs energies and entropies. As long as the vapor behaves as an ideal gas, the standard molar Gibbs energy of vaporization is related to the saturated vapor pressure (p^{sat}) according to Eq. (3):

$$\Delta_1^{\text{g}} G^{\circ} = RT \ln(p^{\text{sat}}/p^{\circ}) \quad (3)$$

where p° is the reference pressure (1 bar).

Root-mean-square deviation (RMS) describes an agreement between the predicted ($f(\text{pred})$) and literature values ($f(\text{lit})$) according to Eq. (4):

$$RMS = \sqrt{\frac{\sum [f(\text{pred}) - f(\text{lit})]^2}{N - 1}} \quad (4)$$

2.2. Relationships between $\Delta_1^{\text{g}} G^{\circ}$ and $\Delta_1^{\text{g}} H^{\circ}$ at 298.15 K in aromatic and aliphatic compounds

Gibbs energy of vaporization is related to the saturated vapor pressure according to Eq. (3), while the vaporization enthalpy defines its temperature dependence. Thus, experimental or reliably evaluated $\Delta_1^{\text{g}} G^{\circ}$ and $\Delta_1^{\text{g}} H^{\circ}$ of organic compounds are required when planning the temperature/pressure regime of distillation, vapor deposition processes, estimating environmental risks associated with volatilization of chemicals [5,22,23].

In Ref. [11], Eq. (5) was shown to describe the relation between $\Delta_1^{\text{g}} G^{\circ}$

and $\Delta_1^{\text{g}} H^{\circ}$ of 180 non-hydrogen-bonded aromatic and short-chain aliphatic non-electrolytes, as well as $\Delta_{\text{solv}} G^{\circ}$ and $\Delta_{\text{solv}} H^{\circ}$ of 406 aromatic and short-chain aliphatic non-electrolytes, di- and triatomic gases in non-hydrogen-bonded solvents at $T_0 = 298.15$ K with $RMS = 1.0$ kJ·mol^{−1}.

$$\Delta_1^{\text{g}} G^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) = 0.660 \Delta_1^{\text{g}} H^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) - 17.0 \quad (5)$$

$\Delta_1^{\text{g}} G^{\circ}$ and $\Delta_1^{\text{g}} H^{\circ}$ of 62 non-hydrogen-bonded long-chain aliphatic compounds ($\Delta_1^{\text{g}} H^{\circ} > 60$ kJ·mol^{−1}) obeyed Eq. (6) with the lower slope ($RMS = 0.6$ kJ·mol^{−1}):

$$\Delta_1^{\text{g}} G^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) = 0.551 \Delta_1^{\text{g}} H^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) - 11.7 \quad (6)$$

Using a weighted sum of Eqs. (4), (5), we could also predict $\Delta_1^{\text{g}} G^{\circ}$ (298.15 K) of 7 low-volatile alkylarenes from their $\Delta_1^{\text{g}} H^{\circ}$ (298.15 K) with a deviation from the literature values below 1.3 kJ·mol^{−1}.

Establishment of the quantitative relationships enables to predict $\Delta_{\text{solv}} G^{\circ}$ ($\Delta_1^{\text{g}} G^{\circ}$) of organic non-electrolytes from $\Delta_{\text{solv}} H^{\circ}$ ($\Delta_1^{\text{g}} H^{\circ}$) and vice versa, which is of particular interest as a quick method for evaluating the vapor pressure in a wide temperature range from a single $p^{\text{sat}} - T$ data point or the vaporization enthalpy at a certain temperature [11].

2.3. Correlations between $\Delta_1^{\text{g}} C_{p,m}$ and $\Delta_1^{\text{g}} H^{\circ}$ at 298.15 K in aromatic and aliphatic compounds

Knowledge of how the vaporization enthalpy varies with temperature is valuable when extrapolating the temperature dependence of the vapor pressure in wide ranges and calculating the heat balances in industrial processes involving the volatilization of fluids [24]. In the fundamental studies of the thermodynamic parameters of phase transitions, knowledge of the temperature dependence of the vaporization enthalpy, which is related to the $\Delta_1^{\text{g}} C_{p,m}$ value, is a cornerstone for the comparative and critical analysis of the data obtained under varying conditions, particularly using the equipment with different temperature and pressure operative ranges [25].

In Ref. [17], we collected the heat capacities of 99 aromatic compounds in the liquid and ideal gas phases and compared the differences with the vaporization enthalpies at $T_0 = 298.15$ K. The linear correlation (Eq. (7)) was obtained with an RMS of 4 J·K^{−1}·mol^{−1}.

$$-\Delta_1^{\text{g}} C_{p,m} / (\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}) = 37.0 + 0.49 \cdot \Delta_1^{\text{g}} H^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) \quad (7)$$

In Refs. [18,19], analogous study was performed on alkanes and aliphatic compounds, and Eq. (8) was proposed for evaluation of their $\Delta_1^{\text{g}} C_{p,m}$ from $\Delta_1^{\text{g}} H^{\circ}$ (298.15 K), also with RMS of 4 J·K^{−1}·mol^{−1}:

$$-\Delta_1^{\text{g}} C_{p,m} / (\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}) = 0.9 + 1.52 \cdot \Delta_1^{\text{g}} H^{\circ} / (\text{kJ} \cdot \text{mol}^{-1}) \quad (8)$$

Alkylaromatic compounds could be described by a weighted sum of Eqs. (7), (8) [20].

Our recent studies showed that $\Delta_1^{\text{g}} H^{\circ}$ (298.15 K) of aromatic [26], aliphatic [27], alkylaromatic [28] compounds could be found from the molecular structure or using solution calorimetry with an accuracy competing with experimental. Therefore, the vaporization enthalpies at the elevated temperature can be adjusted from $T_0 = 298.15$ K using Eqs. (7),(8) according to Kirchhoff's law of Thermochemistry (Eq. (9)):

$$\begin{aligned} \Delta_1^{\text{g}} H^{\circ}(T) &= \Delta_1^{\text{g}} H^{\circ}(T_0) + \int_{T_0}^T \Delta_1^{\text{g}} C_{p,m} dT' \approx \\ &\approx \Delta_1^{\text{g}} H^{\circ}(T_0) + \Delta_1^{\text{g}} C_{p,m} \cdot (T - T_0) \end{aligned} \quad (9)$$

Generally, $\Delta_1^{\text{g}} C_{p,m}$ depends on temperature. This dependence is more pronounced in aliphatic compounds. It can be considered with the

introduction of the empiric correction (see [Supplementary Material](#)).

The validity of Eqs. (7), (8) was independently proven by calculating the temperature dependences of the vaporization enthalpies of >700 aromatic, aliphatic, and alkylaromatic compounds according to Eqs. (7)–(9) and comparing these with the literature $\Delta_1^s H^\circ$ values at elevated temperatures up to 650 K [17–20]. RMS was 1.5 kJ·mol^{−1}, surpassing the previously developed approaches for estimating the temperature dependence of the vaporization enthalpy [25,29,30].

2.4 Expression of $\Delta_1^s G^\circ$ as a function of $\Delta_1^s H^\circ$.

Let us express $\Delta_1^s G^\circ$ as a function of $\Delta_1^s H^\circ$ at $T_0 = 298.15$ K, employing Kirchhoff's law of thermochemistry (Eq. (9)), and correlations between $\Delta_1^s C_{p,m}$ and $\Delta_1^s H^\circ$ (Eq. (2)).

At the normal boiling point (T_b), $\Delta_1^s G^\circ(T_b) = 0$, so Eq. (10) is valid:

$$\Delta_1^s H^\circ(T_b) = T_b \cdot \Delta_1^s S^\circ(T_b) \quad (10)$$

Then T_b can be expressed from Eq. (11):

$$T_b \cdot \Delta_1^s S^\circ(T_b) = \Delta_1^s H^\circ(T_0) + \Delta_1^s C_p \cdot (T_b - T_0) \quad (11)$$

Substituting Eq. (2), one comes to Eq. (12):

$$T_b = \frac{\Delta_1^s H^\circ(T_0) + (c + d \cdot \Delta_1^s H^\circ(T_0)) \cdot T_0}{\Delta_1^s S^\circ(T_b) + c + d \cdot \Delta_1^s H^\circ(T_0)} \quad (12)$$

Eq. (12) correlates the normal boiling point and the vaporization enthalpy at $T_0 = 298.15$ K. This correlation would be different for aliphatics and aromatics due to divergent c and d coefficients. Another variable in Eq. (12) is the vaporization entropy at T_b .

There are many approaches for the estimation of $\Delta_1^s S^\circ(T_b)$ from the molecular structure and physicochemical parameters [24,31–35]. Among the oldest yet effective and simple is Trouton's rule [36], which states that $\Delta_1^s S^\circ(T_b)$ of non-hydrogen-bonded organic non-electrolytes is approximately 88 J·K^{−1}·mol^{−1}. This rule has the shortcomings, described qualitatively and quantitatively previously in the literature [33,37]. But the deviations of $\Delta_1^s S^\circ(T_b)$ from Trouton's rule almost do not further calculations intended to find $\Delta_1^s G^\circ$ from $\Delta_1^s H^\circ$ in the case of organic non-electrolytes with $\Delta_1^s H^\circ(T_0) > 20$ kJ·mol^{−1}, as shown in detail in the [Supplementary Material](#).

Thus, T_b can be calculated from $\Delta_1^s H^\circ(T_0)$, considering that $\Delta_1^s S^\circ(T_b) \approx 88$ J·K^{−1}·mol^{−1}.

From the known T_b and $\Delta_1^s S^\circ(T_b)$, $\Delta_1^s S^\circ(T_0)$ can be found:

$$\begin{aligned} \Delta_1^s S^\circ(T_0) &= \Delta_1^s S^\circ(T_b) + \int_{T_b}^{T_0} \Delta_1^s C_{p,m} \frac{dT}{T} \approx \\ &\approx \Delta_1^s S^\circ(T_b) + \Delta_1^s C_{p,m} \ln(T_0/T_b) \end{aligned} \quad (13)$$

Again, $\Delta_1^s C_p$ can be found according to Eq. (2), while its temperature dependence can be considered as described in [Supplementary Material](#).

Finally, $\Delta_1^s G^\circ(T_0)$ is found from $\Delta_1^s H^\circ(T_0)$ and $\Delta_1^s S^\circ(T_0)$:

$$\Delta_1^s G^\circ(T_0) = \Delta_1^s H^\circ(T_0) - T_0 \Delta_1^s S^\circ(T_0) \quad (14)$$

In case Trouton's rule is used to estimate $\Delta_1^s S^\circ(T_b)$, an analytical expression connecting $\Delta_1^s G^\circ(T_0)$, $\Delta_1^s H^\circ(T_0)$, and c and d coefficients can be written:

$$\begin{aligned} \Delta_1^s G^\circ(T_0) &= \Delta_1^s H^\circ(T_0) - T_0 \cdot \\ &\cdot (\Delta_1^s S^\circ(T_b) + (c + d \cdot \Delta_1^s H^\circ(T_0)) \cdot \\ &\ln\left(\frac{\Delta_1^s H^\circ(T_0)/T_0 + (c + d \cdot \Delta_1^s H^\circ(T_0))}{\Delta_1^s S^\circ(T_b) + c + d \cdot \Delta_1^s H^\circ(T_0)}\right)) \end{aligned} \quad (15)$$

Although it contains $\Delta_1^s H^\circ(T_0)$ under the logarithm, this is asymptotically linear at $\Delta_1^s H^\circ(T_0) > 20$ kJ·mol^{−1}. Eq. (15) is not valid at $\Delta_1^s H^\circ(T_0)$

significantly below 0 kJ·mol^{−1}, since $T_b < 0$ K is anticipated from Eq. (13).

Combining Trouton's rule and Clausius-Clapeyron equation, MacLeod *et al.* [5] also excluded T_b and came to a nearly linear relationship between $\ln p^{\text{sat}}$ and $\Delta_1^s H^\circ$, agreeing with the experimental observations available at that time. However, in Ref. [5] no distinction has been made between aliphatic, aromatic, and hydrogen-bonded compounds, while $\Delta_1^s C_{p,m}$ was assumed to be a constant fraction of $\Delta_1^s S^\circ(T_b)$.

3. Results and discussion

In Fig. 1, the $\Delta_1^s G^\circ$ vs. $\Delta_1^s H^\circ$ dependences predicted for aromatic and aliphatic compounds according to Eq. (15) are depicted by dotted black and red curves, respectively. Black triangles and red circles in Fig. 1 correspond to the literature $\Delta_1^s G^\circ$ and $\Delta_1^s H^\circ$ values of aromatic and aliphatic compounds, respectively, collected in Ref. [11]. Gray crosses, representing alkylarenes, lie between the “aliphatic” and “aromatic” curves. The numerical $\Delta_1^s G^\circ(T_0)$ values calculated according to Eq. (15), corresponding to $\Delta_1^s H^\circ(T_0)$ from 0 to 200 kJ·mol^{−1}, are provided in Table S1. Table S1 additionally contains $\Delta_1^s G^\circ(T_0)$ calculated from Eq. (15) but using other approximations for $\Delta_1^s S^\circ(T_b)$, indicating that the choice of the latter only slightly affects the $\Delta_1^s G^\circ(T_0)$ values (within 1 kJ·mol^{−1}).

In Ref. [11] the $\Delta_1^s G^\circ$ and $\Delta_1^s H^\circ$ values were collected for aromatic and short-chain aliphatic compounds, whose vaporization enthalpies at 298.15 K ranged from 20 to 120 kJ·mol^{−1}. Fitting $\Delta_1^s G^\circ$ and $\Delta_1^s H^\circ$ at $N \rightarrow \infty$ in this range, one can obtain Eq. (16):

$$\Delta_1^s G^\circ / (\text{kJ} \cdot \text{mol}^{-1}) = 0.697 \cdot \Delta_1^s H^\circ / (\text{kJ} \cdot \text{mol}^{-1}) - 18.7 \quad (16)$$

One can evaluate the correspondence between Eq. (16) and the $\Delta_1^s G^\circ$ values calculated according to Eq. (15) by calculating $RMS = 0.24$ kJ·mol^{−1} and $R^2 = 0.99986$ at $N \rightarrow \infty$ by numerical integration (Eq. (17)):

$$RMS = \sqrt{\frac{\int_{x_1}^{x_2} [f(x) - g(x)]^2 dx}{x_2 - x_1}} \quad (17)$$

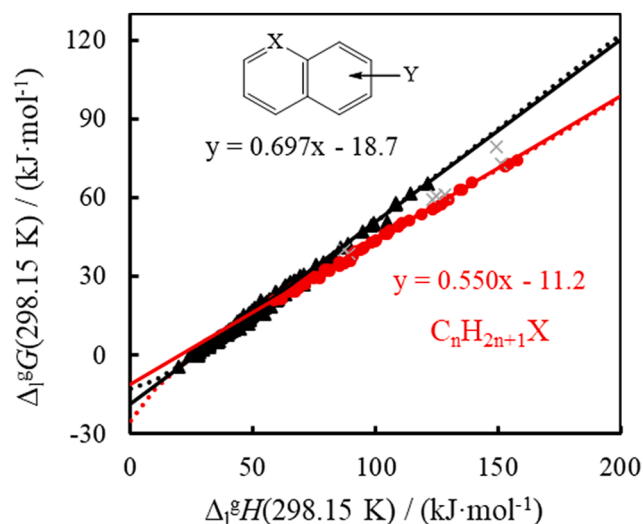


Fig. 1. Graph comparison of the $\Delta_1^s G^\circ$ vs. $\Delta_1^s H^\circ$ correlations predicted from Eq. (15) and the literature $\Delta_1^s G^\circ$ and $\Delta_1^s H^\circ$ values. Black triangles – aromatic and short-chain compounds [11]; red circles – long-chain aliphatic compounds [11]; gray crosses – alkylarenes [11]; dotted black and red curves are described by Eq. (15) with c and d coefficients corresponding to Eqs. (7) (aromatics) and (8) (alkanes), respectively. Solid black and red lines correspond to Eqs. (16) and (18), respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where $f(x)$ corresponds to $\Delta_1^{\text{g}}G^\circ$ calculated from $\Delta_1^{\text{g}}H^\circ$ according to Eq. (16) and $g(x)$ corresponds to $\Delta_1^{\text{g}}G^\circ$ calculated from $\Delta_1^{\text{g}}H^\circ$ according to Eq. (15).

Eq. (16) is depicted by a solid black line in Fig. 1. One can see that both Eqs. (15) and (16) perfectly match the literature $\Delta_1^{\text{g}}G^\circ$ and $\Delta_1^{\text{g}}H^\circ$ values of aromatic and short-aliphatic compounds, while the independently derived coefficients of Eq. (16) agree with those of Eq. (5).

The literature $\Delta_1^{\text{g}}G^\circ$ and $\Delta_1^{\text{g}}H^\circ$ values of long-range aliphatics obeying Eq. (6) lie in the range $60 \text{ kJ}\cdot\text{mol}^{-1} < \Delta_1^{\text{g}}H^\circ < 160 \text{ kJ}\cdot\text{mol}^{-1}$. A linear fit of the data predicted from Eq. (15) leads to the following expression:

$$\Delta_1^{\text{g}}G^\circ / (\text{kJ}\cdot\text{mol}^{-1}) = 0.550 \Delta_1^{\text{g}}H^\circ / (\text{kJ}\cdot\text{mol}^{-1}) - 11.2 \quad (18)$$

From Eq. (17), $RMS = 0.20 \text{ kJ}\cdot\text{mol}^{-1}$ and $R^2 = 0.99984$. Again, the literature $\Delta_1^{\text{g}}G^\circ$ and $\Delta_1^{\text{g}}H^\circ$ values of long-chain aliphatics lie on the curves predicted by Eqs. (15) and (18). The coefficients of Eq. (18) remarkably match those of Eq. (6).

For clarity, the numerical comparison of literature $\Delta_1^{\text{g}}G^\circ$ and $\Delta_1^{\text{g}}H^\circ$ and the $\Delta_1^{\text{g}}G^\circ$ values found from $\Delta_1^{\text{g}}H^\circ$ according to Eqs. (16, 18) for several low-volatile aromatic and aliphatic compounds is provided in Table 1.

Thus, the compensation relationships between $\Delta_1^{\text{g}}G^\circ$ and $\Delta_1^{\text{g}}H^\circ$ of aromatic and aliphatic non-electrolytes at $T_0 = 298.15 \text{ K}$ observed experimentally could be independently derived from the correlations between $\Delta_1^{\text{g}}C_{p,m}$ and $\Delta_1^{\text{g}}H^\circ$, employing Trouton's rule for estimation of the vaporization entropy at the normal boiling point.

It is reasonable to analyze how different choice of $\Delta_1^{\text{g}}C_{p,m}$ and $\Delta_1^{\text{g}}S^\circ(T_b)$ can affect $\Delta_1^{\text{g}}G^\circ$ calculated from $\Delta_1^{\text{g}}H^\circ$ and the form of Eqs. (16, 18). The possible coefficients of $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation obtained under various assumptions are summarized in Table S2.

Generally speaking, increase of the constant $\Delta_1^{\text{g}}S^\circ(T_b)$ value substituted into Eq. (15) leads to decrease of $\Delta_1^{\text{g}}G^\circ$ and intercepts of Eqs. (16, 18) with only slight change of the slope. Particularly, substituting $\Delta_1^{\text{g}}S^\circ(T_b) = 110 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$, which is typical for linear alcohols [33], would shift Eq. (18) by $\sim 5 \text{ kJ}\cdot\text{mol}^{-1}$ downward, agreeing with the previously observed deviation due to solvophobic effect and hydrogen bonding [12]. However, such significant changes of $\Delta_1^{\text{g}}S^\circ(T_b)$ are typical for hydrogen-bonded systems only.

One can anticipate that there may be systematic deviations from Trouton's rule due to molecular size growth, particularly Hildebrand's rule, stating that there is a connection between $\Delta_1^{\text{g}}S^\circ(T_b)$ and $\ln T_b$

Table 1

The literature vaporization enthalpies and Gibbs energies of a series of low-volatile aromatic and aliphatic compounds (columns 2 and 3) and the vaporization Gibbs energies calculated according to Eq. (15).

Compound	$\Delta_1^{\text{g}}H^\circ(\text{lit.})/(\text{kJ}\cdot\text{mol}^{-1})^a$	$\Delta_1^{\text{g}}G^\circ(\text{lit.})/(\text{kJ}\cdot\text{mol}^{-1})^a$	$\Delta_1^{\text{g}}G^\circ(\text{calcd.})/(\text{kJ}\cdot\text{mol}^{-1})$
Aromatics^b			
pyrene	88.5	42.4	43.0
1,2-benzanthracene	98.1	49.4	49.7
3,4-benzopyrene	108.4	57.7	56.9
1,12-benzoperylene	116.8	64.8	62.8
1,2,5,6-dibenzanthracene	121.2	65.2	65.8
Aliphatics^c			
eicosane	100.0	43.8	43.8
heptacosane	134.4	63.0	62.8
7-hexyldocosane	135.2	62.9	63.2
octacontane	139.2	65.7	65.4
11-decylunecosane	152.7	72.0	72.8
11-decylidocosane	157.6	74.5	75.5

^a Ref. [11].

^b $\Delta_1^{\text{g}}G^\circ$ (calcd.) found according to Eq. (15).

^c $\Delta_1^{\text{g}}G^\circ$ (calcd.) found according to Eq. (16).

[31,32]. However, the slight variation of the vaporization entropy, inherent in non-hydrogen-bonded organic compounds, almost does not affect the slopes of linear $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlations derived as described in Sec. 2.3 (see Table S2). The exemplary experimental [38–45] $\Delta_1^{\text{g}}S^\circ(T_b)$ values of compounds with varying T_b and its effect on calculated $\Delta_1^{\text{g}}G^\circ$ (within $1 \text{ kJ}\cdot\text{mol}^{-1}$) are demonstrated in Tables S2 and S3 of the Supplementary Material. What more defines the coefficients of $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation within the experimentally observed variation of $\Delta_1^{\text{g}}S^\circ(T_b)$ of non-hydrogen-bonded organic non-electrolytes is $\Delta_1^{\text{g}}C_{p,m}$ vs. $\Delta_1^{\text{g}}H^\circ$ dependence.

Regarding $\Delta_1^{\text{g}}C_{p,m}$, one may come to the slopes of Eqs. (16, 18) from 1 to 0 if $\Delta_1^{\text{g}}C_{p,m}$ is varied from 0 to infinity (see Table S2). However, together with the weak variation of $\Delta_1^{\text{g}}S^\circ(T_b)$ of non-hydrogen-bonded organic non-electrolytes, the boundaries of the Eqs. (16, 18) coefficients are restricted by the $\Delta_1^{\text{g}}C_{p,m}$ vs. $\Delta_1^{\text{g}}H^\circ$ dependence, which is within the lines given by Eqs. (7) and (8), characterizing aromatic and aliphatic compounds.

One can note from Fig. 1 that Eq. (15) predicts that $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ dependence exhibits curvature at $\Delta_1^{\text{g}}H^\circ < 20 \text{ kJ}\cdot\text{mol}^{-1}$. It contradicts the fact that the linear compensation relationship is observed even at the positive solvation enthalpies (which would correspond to the “negative vaporization enthalpies”) [11]. Actually, Eqs. (7, 8), connecting $\Delta_1^{\text{g}}C_{p,m}$ and $\Delta_1^{\text{g}}H^\circ$ of organic non-electrolytes, were derived in the $\Delta_1^{\text{g}}H^\circ$ range above $20 \text{ kJ}\cdot\text{mol}^{-1}$. $\Delta_1^{\text{g}}C_{p,m}$ of small molecules with $\Delta_1^{\text{g}}H^\circ < 20 \text{ kJ}\cdot\text{mol}^{-1}$ notably drop down, compared with Eqs. (7, 8) [46]. Furthermore, the compounds with $\Delta_1^{\text{g}}H^\circ < 20 \text{ kJ}\cdot\text{mol}^{-1}$ are expected to have boiling points significantly below 298.15 K , and their $\Delta_1^{\text{g}}S^\circ(T_b)$ start deviating from Trouton's rule. If these two features are considered, the curvature is avoided.

4. Conclusion

Although the compensation relationship between the Gibbs energies and enthalpies of solvation and vaporization is long known, it is usually perceived as an extra-thermodynamic correlation with empiric coefficients of unknown origin.

In this paper, we used the recently established $\Delta_1^{\text{g}}C_{p,m}$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation as a tool to study the compensation relationship in non-hydrogen-bonded systems. Employing this correlation together with the basic laws of thermodynamics, we independently found the coefficients of the $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation and investigated the factors defining their values, namely $\Delta_1^{\text{g}}C_{p,m}$ and $\Delta_1^{\text{g}}S^\circ(T_b)$. The variation of $\Delta_1^{\text{g}}S^\circ(T_b)$ mostly affects the intercept of the $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation. However, $\Delta_1^{\text{g}}S^\circ(T_b)$ only slightly varies between non-hydrogen-bonded compounds, so the coefficients of the compensation relationship more depend on $\Delta_1^{\text{g}}C_{p,m}$. Thus, the relationship between $\Delta_1^{\text{g}}C_{p,m}$ and $\Delta_1^{\text{g}}H^\circ$ influences the character of the $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlation. Particularly, the different $\Delta_1^{\text{g}}C_{p,m}$ vs. $\Delta_1^{\text{g}}H^\circ$ relationships in aliphatic and aromatic compounds lead to the $\Delta_1^{\text{g}}G^\circ$ vs. $\Delta_1^{\text{g}}H^\circ$ correlations with the different slopes. The thermodynamic analysis performed enables to see the boundaries within which the $\Delta_1^{\text{g}}G^\circ$ value of a compound with given $\Delta_1^{\text{g}}H^\circ$ can change. Its further application to hydrogen-bonded systems would be of interest.

CRedit authorship contribution statement

Mikhail I. Yagofarov: Conceptualization, Methodology, Writing – original draft. **Boris N. Solomonov:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This paper has been supported by the Kazan Federal University Strategic Academic Leadership Program ('PRIORITY-2030').

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2023.123075>.

References

- [1] L. Liu, Q.-X. Guo, Isokinetic relationship, isoequilibrium relationship, and enthalpy–entropy compensation, *Chem. Rev.* 101 (2001) 673–696.
- [2] I. Barclay, J. Butler, The entropy of solution, *Trans. Faraday Soc.* 34 (1938) 1445–1454.
- [3] R. Krug, W. Hunter, R. Grieger, Enthalpy-entropy compensation. 1. Some fundamental statistical problems associated with the analysis of van't Hoff and Arrhenius data, *J. Phys. Chem.* 80 (1976) 2335–2341.
- [4] A. Pan, T. Biswas, A.K. Rakshit, S.P. Moulik, Enthalpy-entropy compensation (EEC) effect: A revisit, *J. Phys. Chem. B* 119 (2015) 15876–15884.
- [5] M. MacLeod, M. Scheringer, K. Hungerbühler, Estimating Enthalpy of Vaporization from Vapor Pressure Using Trouton's Rule, *Environ. Sci. Technol.* 41 (2007) 2827–2832.
- [6] E. Grunwald, C. Steel, Solvent Reorganization and Thermodynamic Enthalpy-Entropy Compensation, *J. Am. Chem. Soc.* 117 (1995) 5687–5692.
- [7] H.A. Yu, M. Karplus, A thermodynamic analysis of solvation, *J. Chem. Phys.* 89 (1988) 2366–2379.
- [8] A. Vailaya, C. Horváth, Retention in hydrophobic interaction chromatography and dissolution of nonpolar gases in water, *Biophys. Chem.* 62 (1996) 81–93.
- [9] J.E. Leffler, The interpretation of enthalpy and entropy data, *J. Org. Chem.* 31 (1966) 533–537.
- [10] E.S. Rudakov, Thermodynamics of intermolecular interactions, Nauka, Novosibirsk, 1968.
- [11] B.N. Solomonov, M.I. Yagofarov, Compensation relationship in Thermodynamics of solvation and vaporization: features and applications. I. Non-hydrogen-bonded systems, *J. Mol. Liq.* 368 (2022).
- [12] B.N. Solomonov, M.I. Yagofarov, Compensation relationship in thermodynamics of solvation and vaporization: Features and applications. II. Hydrogen-bonded systems, *J. Mol. Liq.* 372 (2023).
- [13] H.S. Frank, Free volume and entropy in condensed systems II. Entropy of vaporization in liquids and the pictorial theory of the liquid state, *J. Chem. Phys.* 13 (1945) 493–507.
- [14] B. Lee, Solvent reorganization contribution to the transfer thermodynamics of small nonpolar molecules, *Biopolymers* 31 (1991) 993–1008.
- [15] I.A. Sedov, M.A. Stolov, B.N. Solomonov, Solvophobic effects and relationships between the Gibbs energy and enthalpy for the solvation process, *J. Phys. Org. Chem.* 24 (2011) 1088–1094.
- [16] M.D. Borisover, F.D. Baitalov, B.N. Solomonov, Evaluation of the contribution to hydration of nonelectrolytes from the hydrophobic effect, *J. Sol. Chem.* 24 (1995) 579–586.
- [17] M.I. Yagofarov, D.N. Bolmatenkov, B.N. Solomonov, Relationship between the vaporization enthalpies of aromatic compounds and the difference between liquid and ideal gas heat capacities, *J. Chem. Thermodyn.* 158 (2021), 106443.
- [18] M.I. Yagofarov, A.A. Sokolov, D.N. Bolmatenkov, B.N. Solomonov, Relationship between the heat capacity change on vaporization of normal and branched alkanes and the vaporization enthalpy and its prediction as a function of temperature, *J. Chem. Thermodyn.* 163 (2021), 106586.
- [19] A.A. Sokolov, D.N. Bolmatenkov, M.I. Yagofarov, I.S. Balakhontsev, B. N. Solomonov, Estimation of the temperature dependence of the vaporization enthalpies of monofunctional aliphatic compounds, *Fluid Phase Equilib.* 553 (2022), 113304.
- [20] D.N. Bolmatenkov, M.I. Yagofarov, A.A. Notfullin, B.N. Solomonov, Calculation of the vaporization enthalpies of alkylaromatic hydrocarbons as a function of temperature from their molecular structure, *Fluid Phase Equilib.* 113303 (2022).
- [21] B.N. Solomonov, V.B. Novikov, Solution calorimetry of organic nonelectrolytes as a tool for investigation of intermolecular interactions, *J. Phys. Org. Chem.* 21 (2008) 2–13.
- [22] K.-U. Goss, R.P. Schwarzenbach, Empirical prediction of heats of vaporization and heats of adsorption of organic compounds, *Environ. Sci. Technol.* 33 (1999) 3390–3393.
- [23] M. Shtein, H.F. Gossenberger, J.B. Benziger, S.R. Forrest, Material transport regimes and mechanisms for growth of molecular organic thin films using low-pressure organic vapor phase deposition, *J. Appl. Phys.* 89 (2001) 1470–1476.
- [24] Z. Kolská, V. Růžicka, R. Gani, Estimation of the enthalpy of vaporization and the entropy of vaporization for pure organic compounds at 298.15 K and at normal boiling temperature by a group contribution method, *Ind. Eng. Chem. Res.* 44 (2005) 8436–8454.
- [25] J.S. Chickos, S. Hosseini, D.G. Hesse, J.F. Liebman, Heat capacity corrections to a standard state: a comparison of new and some literature methods for organic liquids and solids, *Struct. Chem.* 4 (1993) 271–278.
- [26] B.N. Solomonov, M.I. Yagofarov, An approach for the calculation of vaporization enthalpies of aromatic and heteroaromatic compounds at 298.15 K applicable to supercooled liquids, *J. Mol. Liq.* 319 (2020).
- [27] R.N. Nagrimanov, A.A. Samatov, B.N. Solomonov, Additive scheme of solvation enthalpy for linear, cyclic and branched-chain aliphatic compounds at 298.15 K, *J. Mol. Liq.* 292 (2019).
- [28] B.N. Solomonov, M.I. Yagofarov, R.N. Nagrimanov, Additivity of vaporization enthalpy: group and molecular contributions exemplified by alkylaromatic compounds and their derivatives, *J. Mol. Liq.* 342 (2021), 117472.
- [29] D. Dalmazzone, A. Salmon, S. Guella, A second order group contribution method for the prediction of critical temperatures and enthalpies of vaporization of organic compounds, *Fluid Phase Equilib.* 242 (2006) 29–42.
- [30] N. Cohen, S.W. Benson, The Thermochemistry of Alkanes and Cycloalkanes, in: *Alkanes and Cycloalkanes* (1992), 1992, pp. 215–287.
- [31] J.H. Hildebrand, The entropy of vaporization as a means of distinguishing normal liquids, *J. Am. Chem. Soc.* 37 (1915) 970–978.
- [32] D. Everett, 519. Some correlations between thermodynamic properties and the structure of liquids, *J. Chem. Soc.* (1960) 2566–2573.
- [33] P.B. Myrdal, J.F. Krzyzaniak, S.H. Yalkowsky, Modified Trouton's rule for predicting the entropy of boiling, *Ind. Eng. Chem. Res.* 35 (1996) 1788–1792.
- [34] D. Hoshino, K. Nagahama, M. Hirata, Prediction of the entropy of vaporization at the normal boiling point by the group contribution method, *Ind. Eng. Chem. Fund.* 22 (1983) 430–433.
- [35] L. Zhao, P. Li, S.H. Yalkowsky, Predicting the entropy of boiling for organic compounds, *J. Chem. Inf. Comput. Sci.* 39 (1999) 1112–1116.
- [36] F.T. Trouton, *Philos. Mag.* 5 (1884) 54–58.
- [37] J.A. Green, S.J. Irudayam, R.H. Hinchman, Molecular interpretation of Trouton's and Hildebrand's rules for the entropy of vaporization of a liquid, *The, J. Chem. Thermodyn.* 43 (2011) 868–872.
- [38] V. Majer, V. Svoboda, Enthalpies of vaporization of organic compounds: A critical review and data compilation, Blackwell Scientific Publications, Oxford, 1985.
- [39] V. Majer, V. Svoboda, M. Lencka, Enthalpies of vaporization and cohesive energies of dimethylpyridines and trimethylpyridines, *J. Chem. Thermodyn.* 17 (1985) 365–370.
- [40] W. Steele, R. Chirico, S. Knipmeyer, A. Nguyen, The thermodynamic properties of benzothiazole and benzoxazole, *J. Chem. Thermodyn.* 24 (1992) 499–529.
- [41] R. Chirico, S. Knipmeyer, A. Nguyen, W. Steele, The thermodynamic properties to the temperature 700 K of naphthalene and of 2, 7-dimethylnaphthalene, *J. Chem. Thermodyn.* 25 (1993) 1461–1494.
- [42] R.D. Chirico, S.E. Knipmeyer, A. Nguyen, W.V. Steele, The thermodynamic properties of benzo[b]thiophene, *J. Chem. Thermodyn.* 23 (1991) 759–779.
- [43] W.V. Steele, R.D. Chirico, S.E. Knipmeyer, A. Nguyen, The thermodynamic properties of 9-methylcarbazole and of 1,2,3,4-tetrahydro-9-methylcarbazole, *J. Chem. Thermodyn.* 24 (1992) 245–271.
- [44] R.D. Chirico, A.F. Kazakov, W.V. Steele, Thermodynamic properties of 9-fluorenone: Mutual validation of experimental and computational results, *J. Chem. Thermodyn.* 54 (2012) 278–287.
- [45] R.D. Chirico, A.F. Kazakov, W.V. Steele, Thermodynamic properties of three-ring aza-aromatics. 1. Experimental results for phenazine and acridine, and mutual validation of experiments and computational methods, *J. Chem. Thermodyn.* 42 (2010) 571–580.
- [46] M.I. Yagofarov, A.A. Sokolov, B.N. Solomonov, Comparison of isochoric and isobaric heat capacities of liquid organic non-electrolytes and their vaporization enthalpies, *J. Mol. Liq.* 116886 (2021).